

Lab 4 - Logic Equivalence Check (LEC)

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Objective

To understand the basics of Conformal LEC setup, and debug basic non-equivalences for the Sequential Multiplier design.

Task 1: LEC for Sequential Multiplier System

Objective

Perform a standard Logic Equivalence Check between the original RTL and the synthesized gate-level netlist to confirm that the synthesis tool correctly mapped the logic.

Implementation

The setup script (`multiplier_lec.do`) was created to read the library, read the Golden RTL files (`controller.v`, `datapath.v`, `top.v`), and read the Revised gate-level netlist (`top_gates.v`) generated by Genus.

```
1 set log file multiplier_lec.log -replace
2 read library ../lib/LIB/slow_vdd1v0_basiccells.lib -liberty -both
3 read design ../rtl/controller.v ../rtl/datapath.v ../rtl/top.v -verilog
  -golden
4 read design ../work/top_gates.v -verilog -revised
5 set system mode lec
6 add compared point -all
7 compare
8 report verification -verbose
```

Listing 1: `multiplier_lec.do`

Verification Result

The Conformal LEC tool was run using the above script. The comparison successfully mapped all 27 DFFs and primary outputs. The verification report confirms 0 Non-Equivalent points.

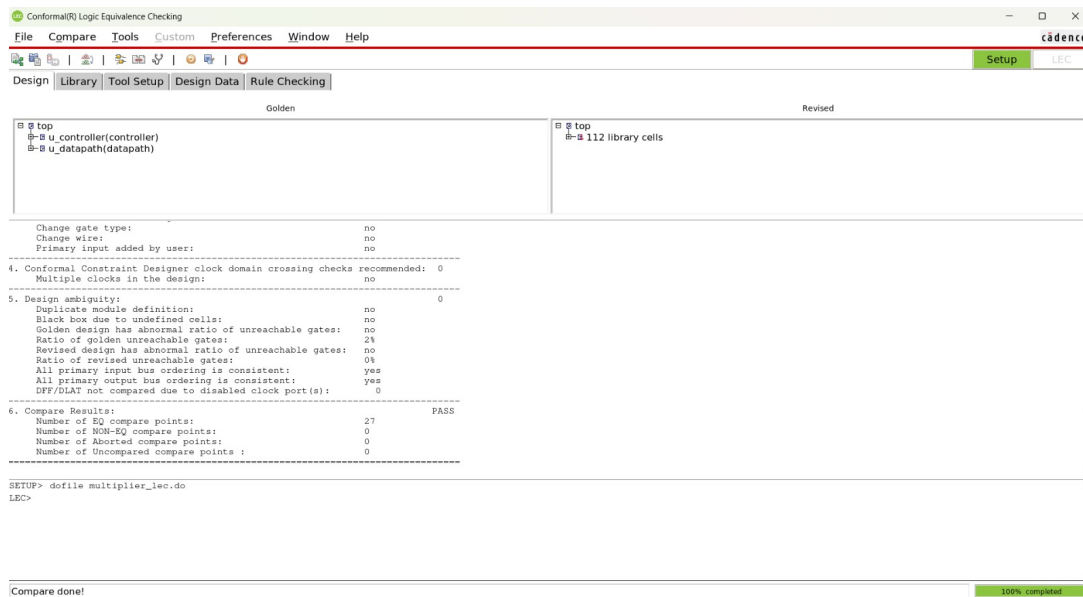


Figure 1: Task 1: LEC PASS with 0 Non-Equivalent points

Task 2: LEC for Sequential Multiplier with Clock Gating

Objective

Enable automatic clock gating insertion during synthesis for power savings, generate a report of deleted sequential elements, and successfully run LEC on the clock-gated netlist by providing the appropriate flattening constraint.

Synthesis Modification

The Genus synthesis script was modified to include clock gating. The following line was added before reading the Verilog files:

```
1 set_db / .lp_insert_clock_gating true
```

Additionally, a report was generated to track the flip-flops that were optimized out by the clock gating logic:

```
1 report_sequential -deleted > work_cg/deleted_flops.rpt
```

LEC without Constraint (Demonstration of Failure)

When running standard LEC on the clock-gated netlist, the verification fails because Conformal cannot map the new clock gating cells (RC_CG_MOD) present in the Revised netlist to the Golden RTL.

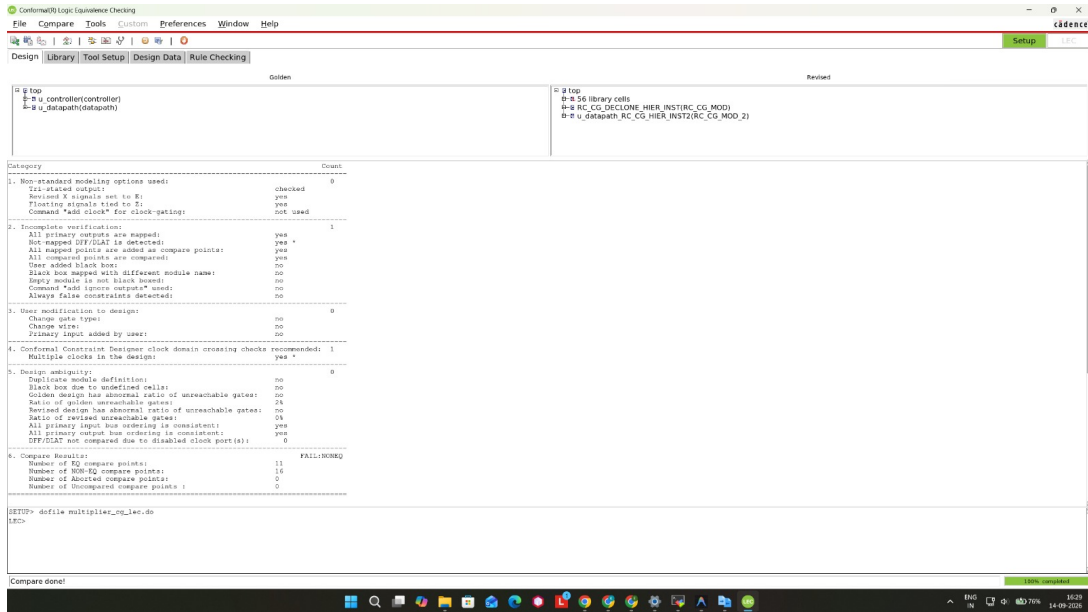


Figure 2: Task 2: LEC Failure (16 Non-Equivalent points) due to unmodeled Clock Gating

LEC with Constraint (Fix)

To instruct the tool to correctly recognize and mathematically flatten the clock gating logic, the following constraint was added to the .do file before switching to LEC mode:

```
1 set flatten model -gated_clock
```

After applying this constraint, Conformal was able to correctly compare the logic and verify equivalence.

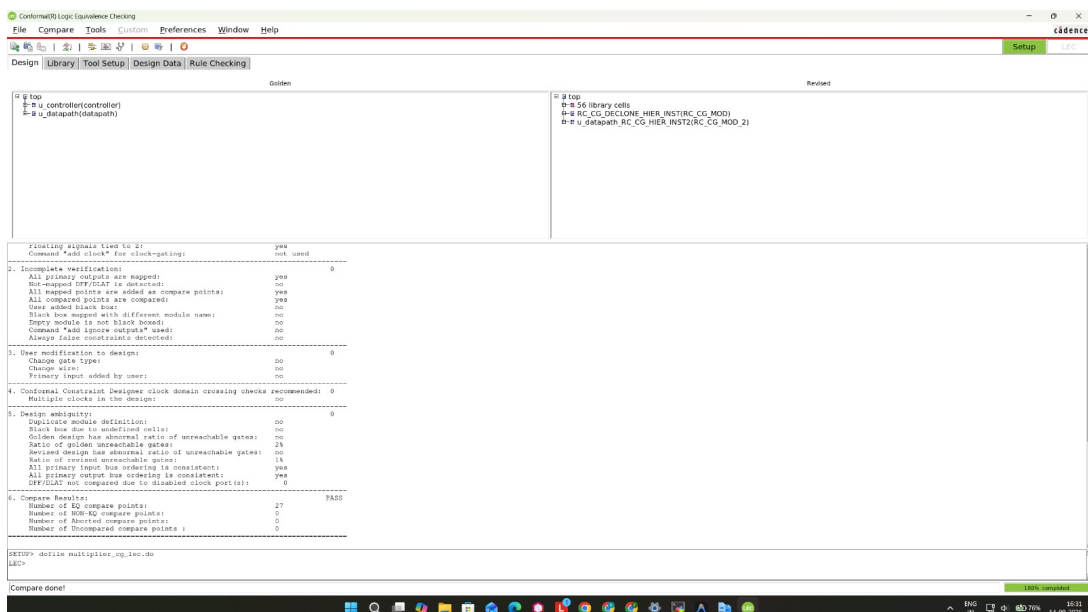


Figure 3: Task 2: LEC PASS with flattened clock gating models

Task 3: LEC Run with a Netlist Bug

Objective

To intentionally inject a bug into the gate-level netlist and utilize Conformal LEC's graphical debug tools (Mapping Manager and Diagnosis Manager) to identify the root cause of the non-equivalence.

Bug Injection

A manual edit was made to the `top_gates_buggy.v` netlist. The input to an inverter (g1072) was incorrectly re-routed from `accumulated_prod[5]` to `accumulated_prod[4]`.

Debugging Process

Upon running LEC, the tool reported 3 Non-Equivalent compare points. Using the GUI, the following debugging steps were taken:

1. Opened the **Mapping Manager** to filter and identify the specific Non-Equivalent DFFs.
2. Right-clicked the failing point `accumulated_prod_reg[5]` and selected **Diagnose**.
3. In the **Diagnosis Manager**, Conformal's mathematical solver analyzed the logic cone and correctly identified `INVX1 g1072` as the primary Error Candidate with a probability of 1.00.
4. The **Schematics** view was utilized to visually trace the boolean logic error backward from the DFF input, confirming that the Revised logic was pulling from bit [4] while the Golden reference was pulling from bit [5].

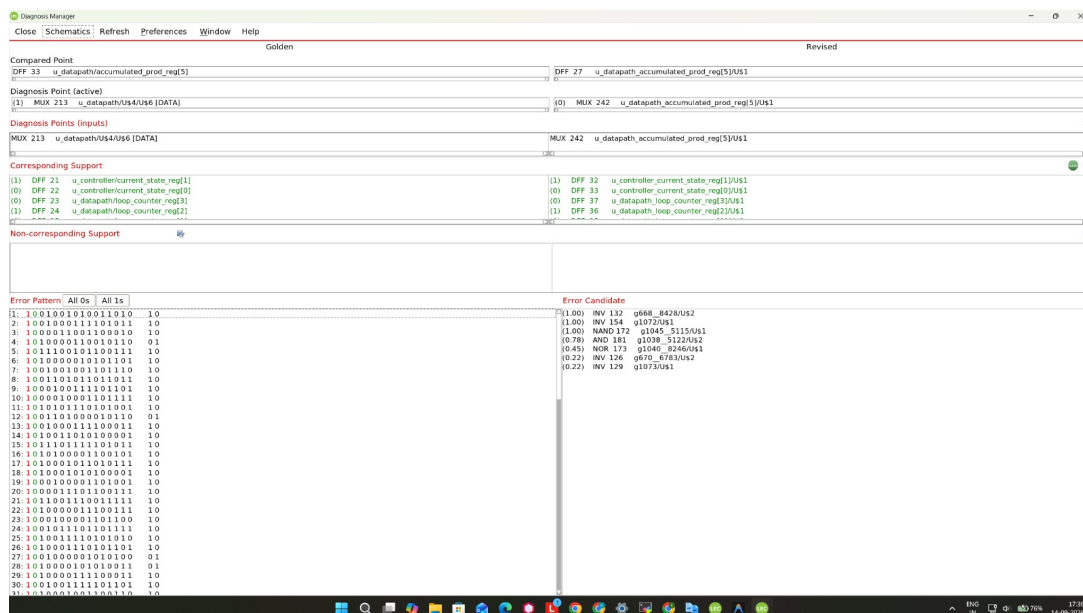


Figure 4: Task 3: Diagnosis Manager identifying g1072 with 1.00 probability

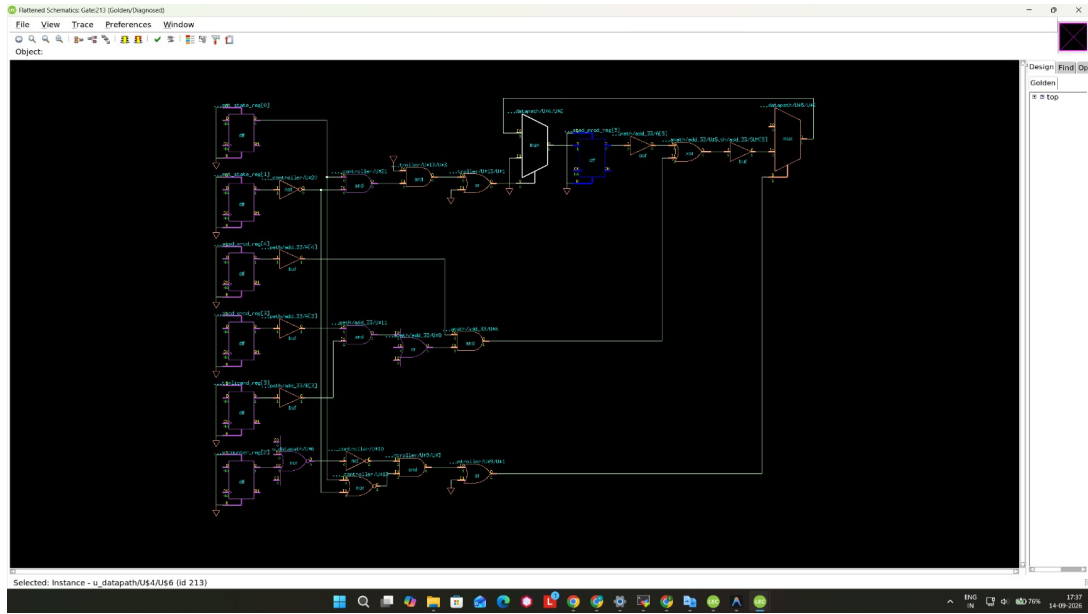


Figure 5: Task 3: Golden flattened schematic at `accumulated_prod_reg[5]`, driven correctly from bit [5]

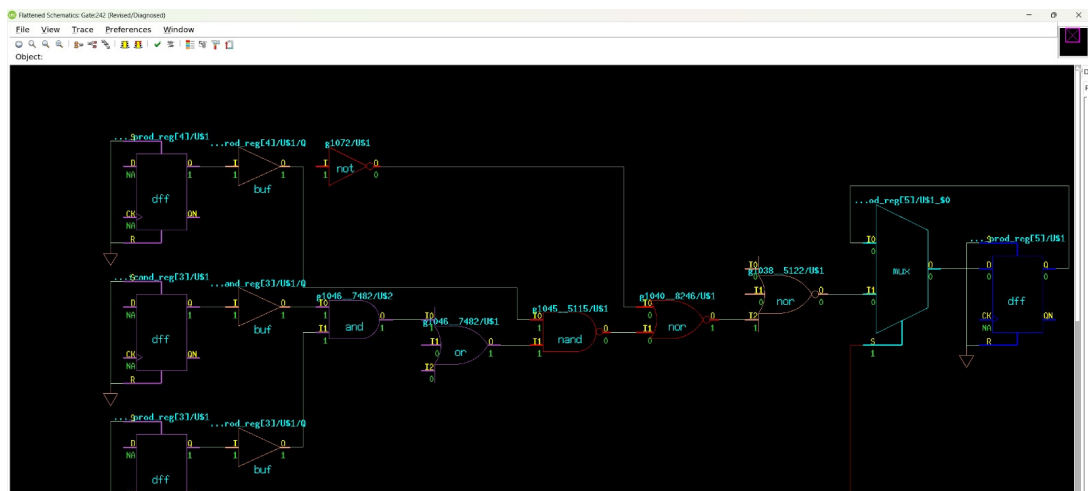


Figure 6: Task 3: Revised flattened schematic isolating the fault region around `g1072`

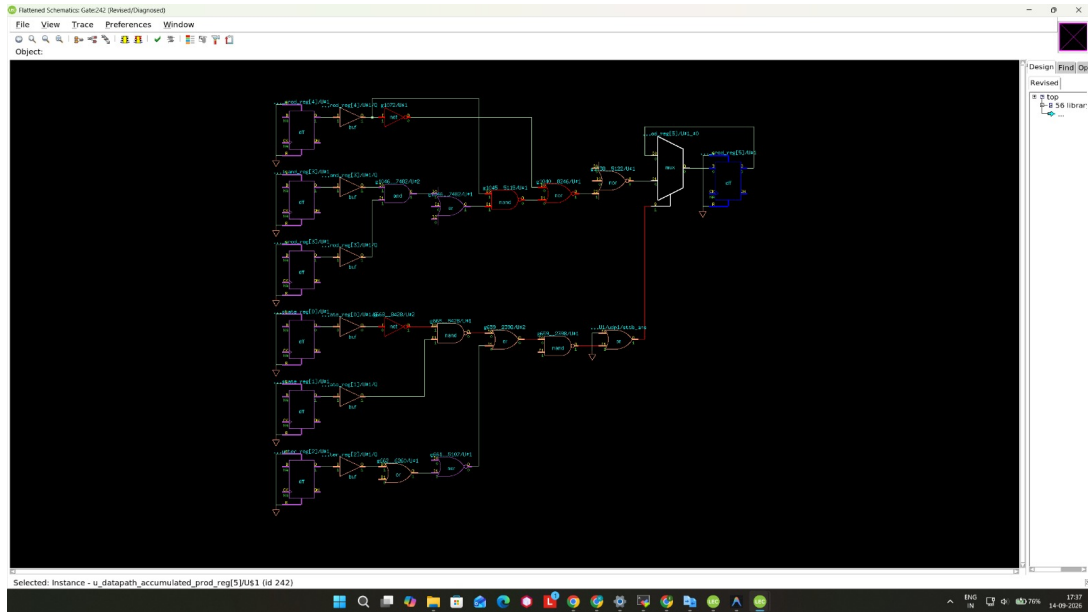


Figure 7: Task 3: Flattened Schematics tracing the error path backward to g1072, showing the incorrect re-route from bit [4]

Bug Fix and Verification

The buggy netlist was corrected by changing the wire mapping back to `accumulated_prod[5]`. A final LEC run confirmed the fix.

Command	*Add clock* for clock-gating	not used
2. Incomplete verification:		0
All primary outputs are mapped:	yes	
Not-mapped DFF/DLUT is detected:	no	
All mapped points are added as compare points:	yes	
All compare points are compared:	yes	
Don't added black box:	no	
Black box mapped with different module name:	no	
Empty module is not black boxed:	no	
Command "Add ignore outputs" used:	no	
Always false constraints detected:	no	
3. User modification to design:		0
Change gate type:	no	
Change wire:	no	
Primary input added by user:	no	
4. Conformal Constraint Designer clock domain crossing checks recommended:	0	
Multiple clocks in the design:	no	
5. Design ambiguity:		0
Duplicate module definition:	no	
Black box due to undefined cells:	no	
Golden design has abnormal ratio of unreachable gates:	no	
Ratio of golden unreachable gates:	23	
Revised design has abnormal ratio of unreachable gates:	no	
Ratio of revised unreachable gates:	18	
All primary input bus ordering is consistent:	yes	
All primary output bus ordering is consistent:	yes	
DFF/DLUT not compared due to disabled clock port(s):	0	
6. Compare Results:		PASS
Number of EQ compare points:	27	
Number of XOR-EQ compare points:	0	
Number of XORed compare points:	0	
Number of Uncompared compare points:	0	

Figure 8: Task 3: Final LEC PASS after fixing the netlist bug